

Quantum mechanics in momentum space: An illustration

M. Lieber

Department of Physics

University of Arkansas

Fayetteville, Arkansas 72701

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The formulation of quantum-mechanical problems in momentum space, which is largely ignored in traditional courses but is used in research, is discussed. In particular, the formulation of scattering problems, for which the momentum representation is quite natural, is neither generally well known nor obvious. This is illustrated by solving a simple problem, the one-dimensional delta function potential, for which the bound state and continuum wave function are found.

I. INTRODUCTION

In traditional quantum mechanics courses,^{1,2} students are exposed to a wide variety of problems, all of which are defined in configuration space (x representation). While momentum space (p representation) is generally introduced in a formal way,³ it is rarely used in the solution of problems.⁴ Any momentum space functions which are needed are generally derived by solving the problem in the x representation and then Fourier transforming the result. This is no doubt because the potential function in configuration space becomes an integral operator in momentum space and the Schrödinger equation becomes an integral equation. However, for many problems the momentum representation is actually more natural. This is particularly true in scattering problems, where one is interested in calculating the probability amplitude for the change in direction of the momentum vector. In fact, in the usual treatment of potential scattering found in textbooks,⁵ the Schrödinger equation is converted into an integral equation by means of the free particle Green's function, and the latter is found by means of Fourier transformation. This technique is in effect utilizing the simple form of the free Green's function in momentum space, where the kinetic energy operator is diagonal. Since the x -space and p -space equations are both integral equations, the advantage of the former would appear to be lost. Other arguments for using the p representation have been given by Dirac in his classic text.⁶

It is the purpose of this paper to illustrate the solution of the Schrödinger equation in momentum space by means of a simple example for which the configuration space solution is relatively familiar: the one-dimensional motion of a particle in a delta function potential.⁷ While this potential may not be physically realistic, most of the

characteristics of the Schrödinger equation may be illustrated by its use, and so it is of pedagogical value.

II. SCHRÖDINGER EQUATION IN MOMENTUM SPACE

We begin with the Schrödinger equation in the x representation

$$(E - \hat{p}^2/2m)\psi(x) = V(x)\psi(x), \quad (1)$$

where $\hat{p}$ represents the operator $-i\hbar d/dx$. The momentum space wave function $\phi(p)$ is defined by

$$\phi(p) = (2\pi\hbar)^{-1/2} \int_{-\infty}^{\infty} \exp(-ipx/\hbar) \psi(x) dx \quad (2)$$

with the inverse relation

$$\psi(x) = (2\pi\hbar)^{-1/2} \int_{-\infty}^{\infty} \exp(ipx/\hbar) \phi(p) dp. \quad (3)$$

The function $\phi(p)$ plays the same role in the p representation that $\psi(x)$ plays in x space; for example, the probability that a measurement of the momentum will fall between p and $p + dp$ is given by $|\phi(p)|^2 dp$. In terms of $\phi(p)$ the Schrödinger equation, Eq. (1), becomes an integral equation:

$$(E - p^2/2m)\phi(p) = (2\pi\hbar)^{-1/2} \int_{-\infty}^{\infty} \tilde{V}(p - p') \phi(p') dp'. \quad (4)$$

In this equation p is merely a number (the kinetic energy operator, like the momentum, is diagonal in the p representation), while $\tilde{V}(p - p')$ is, apart from the occurrences of $\hbar$, the Fourier transform of $V(x)$:

$$\tilde{V}(p) = (2\pi\hbar)^{-1/2} \int_{-\infty}^{\infty} \exp(-ipx/\hbar) V(x) dx. \quad (5)$$

In this paper we shall take for illustrative purposes the potential

$$V(x) = c\delta(x), \quad (6)$$

where c is a constant of dimension energy times length. For this potential we have

$$\tilde{V}(p) = (2\pi\hbar)^{-1/2} c, \quad (7)$$

and Eq. (4) takes the particularly simple form

$$(E - p^2/2m)\phi(p) = (c/2\pi\hbar) \int_{-\infty}^{\infty} \phi(p) dp. \quad (8)$$

III. BOUND STATE SOLUTION

Let us define a constant α , unknown at present, by

$$\alpha = (mc/\pi\hbar) \int_{-\infty}^{\infty} \phi(p) dp. \quad (9)$$

Since we are looking for bound states, for which $E < 0$, it will be convenient to set

$$\beta = (-2mE)^{1/2} \quad (10)$$

or, conversely,

$$E = -\beta^2/2m. \quad (11)$$

Then Eq. (8) can be written in the form

$$(p^2 + \beta^2)\phi(p) = -\alpha, \quad (12)$$

so that

$$\phi(p) = -\alpha/(p^2 + \beta^2). \quad (13)$$

If we now substitute this into Eq. (9), we find

$$\alpha = -(mc/\hbar\beta)\alpha, \quad (14)$$

which is possible only if

$$\beta = -mc/\hbar. \quad (15)$$

Since β is the positive square root, this condition can be satisfied only if $c < 0$ (attractive potential). In that case we have this result⁷ from Eq. (11):

$$E = -mc^2/2\hbar^2. \quad (16)$$

The corresponding bound state wave function is given by Eq. (13), where β has the value of (15) and α remains

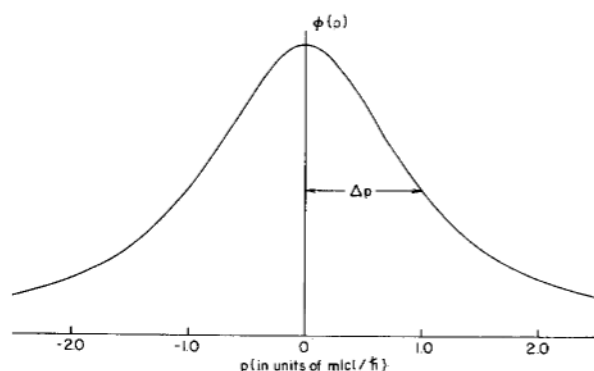


Fig. 1. Bound state wave function in momentum space. We have indicated the "half-width" Δp referred to in the text, one-half the distance between points $\pm p$, where $\phi(p) = \frac{1}{2}\phi(0)$.

for purposes of normalization. The resulting wave function,

$$\phi(p) = \left(\frac{2}{\pi}\right)^{1/2} \frac{(m|c|/\hbar)^{3/2}}{p^2 + (m|c|/\hbar)^2} \quad (17)$$

is sketched in Fig. 1. Note that $m|c|/\hbar$ represents the "half-width" of the Lorentzian-shaped wave function (half-width at half-maximum). While a measurement of momentum may give any value, the most likely value will be zero, with most values falling between $-m|c|/\hbar$ and $+m|c|/\hbar$.

By inserting Eq. (17) into Eq. (3) and evaluating the resulting expression (by contour integration, treating the cases of $x > 0$ and $x < 0$ separately), we obtain the configuration space wave function:

$$\psi(x) = (m|c|/\hbar^2)^{1/2} \exp(-m|cx|/\hbar^2). \quad (18)$$

This result is usually obtained by using the boundary conditions at $x = 0$: that is, continuity of $\psi(x)$ but discontinuity of its slope, $d\psi/dx$ (see Ref. 7). The wave function $\psi(x)$ is sketched in Fig. 2. The "half-width" in this case, defined for simplicity as half the distance between the points where $\psi(x)$ falls to $1/e$ of its maximum, is $\hbar^2/m|c|$. While the "widths" of $\phi(p)$ and $\psi(x)$ are not defined in precisely the requisite manner, their product is $\hbar$, in agreement with the indeterminacy principle.

It is an interesting exercise to obtain the above results as a limiting case of a square-well potential: $V(x) = -V_0$ for $|x| < a$; $V(x) = 0$ otherwise. The requisite limit is $V_0 \rightarrow \infty$, $a \rightarrow 0$ with $2V_0a = |c|$ held finite.

IV. SCATTERING PROBLEM

The scattering problem is rather more complicated in any representation. In configuration space one seeks solutions to Eq. (1) which have the form, for large $|x|$, of "plane wave" plus "outgoing wave." More explicitly, let us take a plane wave representing an incident particle with momentum p_0 moving to the right:

$$\psi_0(x) = (2\pi\hbar)^{-1/2} \exp(ip_0x/\hbar). \quad (19)$$

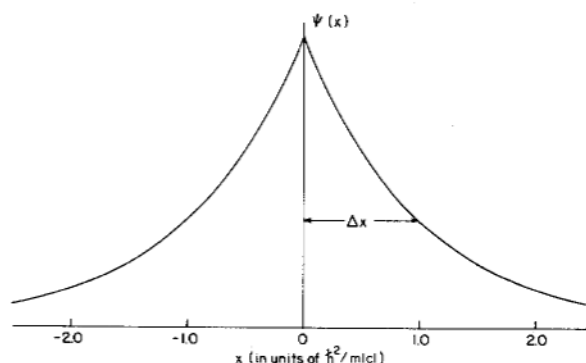


Fig. 2. Bound state wave function in configuration space. This is essentially the Fourier transform of the curve in Fig. 1. In this case we have defined the "half-width" as one-half the distance between points $\pm x$, where $\psi(x) = (1/e)\psi(0)$.

The normalization constant has been chosen so that

$$\int_{-\infty}^{\infty} \psi_{p_0'}(x)^* \psi_{p_0}(x) dx = \delta(p_0 - p_0') \quad (20)$$

(which is Merzbacher's p normalization⁸). The fact that $|\psi_0|^2$ is independent of x , so that the particle is totally unlocalizable, is a consequence of the precise specification of its momentum, in accordance with the indeterminacy principle. The connection between this obviously unphysical formulation and the experimental situation is well described in Ref. 2.

We seek a solution to Eq. (1), $\psi(x)$, with the property that its scattered part, namely $\psi = \psi - \psi_0$, is purely outgoing for large $|x|$. This means that, as $|x| \rightarrow \infty$, $\psi(x)$ has the asymptotic form $\exp(ip_0|x|/\hbar)$, which is a plane wave moving to the right for $x \rightarrow +\infty$ and to the left for $x \rightarrow -\infty$.

In momentum space the plane wave (19) becomes, upon substitution into Eq. (2),

$$\phi_0(p) = \delta(p - p_0). \quad (21)$$

This wave function is normalized according to

$$\begin{aligned} \int_{-\infty}^{\infty} \phi_{p_0'}(p)^* \phi_{p_0}(p) dp &= \int_{-\infty}^{\infty} \delta(p - p_0') \delta(p - p_0) dp \\ &= \delta(p_0 - p_0'), \end{aligned} \quad (22)$$

in agreement with Eq. (20) and the Parseval identity.

The appearance of a *generalized* function ϕ_0 instead of an ordinary function is the momentum space counterpart to the lack of normalizability of ψ_0 in the usual square-integral sense. This problem does not occur for bound state functions, which are normalizable, but evidently must be a general feature of continuum solutions in the p representation and cannot be avoided except by the use of cumbersome wave packets.

The plane waves, Eqs. (19) and (21), satisfy the respective free particle Schrödinger equations,

$$(E - \hat{p}^2/2m)\psi_0(x) = 0 \quad (23a)$$

and

$$(E - p^2/2m)\phi_0(p) = 0, \quad (23b)$$

where the energy is given by

$$E = p_0^2/2m. \quad (24)$$

Equation (23a) is a differential equation, whereas, by contrast, Eq. (23b) is merely algebraic, a consequence of the rule⁹

$$y\delta(y) = 0, \quad (25)$$

where $y = p_0 - p$ in the present case.

How are we to formulate the "outgoing wave" condition in momentum space? We seek a solution to Eq. (4), $\phi(p)$, with the property that the scattered part, $\bar{\phi} = \phi - \phi_0$, has only outgoing waves. Dirac⁶ has shown, in the three-dimensional case, that ϕ will be purely outgoing if it contains the factor

$$P[(E - p^2/2m)^{-1}] - i\pi\delta(E - p^2/2m), \quad (26)$$

where P means the "Cauchy principal value." This specification determines the nature of the singularity at $p^2 = 2mE$ in performing any subsequent integration over p . Such integrations are always implicit when generalized functions are used. The proof that the occurrence of (26) as a factor in ϕ (the other factor is a smooth function of p) does indeed correspond to the outgoing wave condition is rather technical and hence is relegated to the Appendix. Here we will note that, using the identity¹⁰

$$P(1/y) - i\pi\delta(y) = \lim_{\epsilon \rightarrow 0^+} [(y + i\epsilon)^{-1}], \quad (27)$$

the Dirac factor can be cast in the form

$$\lim_{\epsilon \rightarrow 0^+} [(E + i\epsilon - p^2/2m)^{-1}], \quad (28)$$

which is just the standard trick⁵ for displacing the poles in the derivation of the free Green's function with outgoing waves. In fact, apart from a momentum-conserving delta function, (28) is the free Green's function in the p representation. In expressions such as (27) and (28) the ϵ limit is intended to be taken *after* the subsequent integrations, which are implied by the generalized functions.

Let us now turn to the solution of our scattering problem with the delta function potential. We begin with Eq. (8) and make the same definition, Eq. (9), used in the bound state problem, with the resulting equation

$$(p_0^2 - p^2)\phi(p) = \alpha. \quad (29)$$

Unlike the bound state case we cannot simply divide by $p_0^2 - p^2$ because this factor vanishes when $p = \pm p_0$, whereas in the previous case, Eq. (12), the factor $p^2 + \beta^2$ was always positive. We must specify the treatment at these singularities. If we define the reciprocal of $p_0^2 - p^2$ to be the principal value, as is customary in the theory of generalized functions, the general solution to Eq. (29) can be written

$$\phi(p) = A\delta(p - p_0) + B\delta(p + p_0) + \alpha P[(p_0^2 - p^2)^{-1}]. \quad (30)$$

The two delta function terms are permissible because, according to the rule (25), they are each annihilated upon application of $p_0^2 - p^2$. As in the bound state case, we now substitute Eq. (30) into Eq. (9). The principal value integral vanishes, and we get simply

$$\alpha = (mc/\pi\hbar)(A + B), \quad (31)$$

which is to be contrasted with the consistency condition, Eq. (14), that occurred at this stage in the bound state discussion.

We have yet to impose the outgoing wave condition. Our solution must in general have the form

$$\phi(p) = \delta(p - p_0) + \{2mP[(p_0^2 - p^2)^{-1}] - i(m\pi/p_0)[\delta(p + p_0) + \delta(p - p_0)]\}f(p), \quad (32)$$

where $f(p)$ is a smooth function determined by the potential. The first term on the right-hand side of Eq. (32) is $\phi_0(p)$, Eq. (21), while the rest is $\bar{\phi}$; the factor in braces is simply Dirac's factor (26), rewritten by using the identity¹¹

$$\delta(p_0^2 - p^2) = [\delta(p + p_0) + \delta(p - p_0)]/2p_0. \quad (33)$$

Comparing the principal value terms in Eqs. (32) and (30) we see that, in our case, $f(p)$ is a constant independent of p :

$$f(p) = \alpha/2m \equiv \lambda \quad (34)$$

or, by using Eq. (31),

$$\lambda = (c/2\pi\hbar)(A + B). \quad (35)$$

Next, we compare the coefficients of $\delta(p + p_0)$ and $\delta(p - p_0)$ in Eqs. (32) and (30). We find

$$A = 1 - im\pi\lambda/p_0, \\ B = -im\pi\lambda/p_0,$$

so that, inserting these into Eq. (35), we have

$$\lambda = (c/2\pi\hbar)(1 - 2im\pi\lambda/p_0).$$

This equation may be solved for λ :

$$\lambda = \frac{cp_0/2\pi}{\hbar p_0 + imc}. \quad (36)$$

Thus, our solution is, finally,

$$\phi(p) = \delta(p - p_0) + \lambda \{P[(E - p^2/2m)^{-1}] - i\pi\delta(E - p^2/2m)\} \quad (37)$$

with λ given by Eq. (36). If the Dirac factor is used in the form of (28), the Green's function form, then λ is identified as the conventional scattering amplitude. Expression (37) [cf. (32)] contains both $\delta(p - p_0)$, representing waves traveling to the right, and $\delta(p + p_0)$, representing the reflected waves traveling to the left. The principal value term serves to cancel the wave traveling in the wrong direction in each region of x space, as will be seen

immediately below. It is indicative of plane waves which are only semi-infinite in extent.

The reader is invited to transform this solution back to the x representation, using Eq. (3). The result is

$$\psi(x) = (2\pi\hbar)^{-1/2} \left[\exp\left(\frac{ip_0x}{\hbar}\right) + \frac{mc}{i\hbar p_0 - mc} \exp\left(\frac{ip_0|x|}{\hbar}\right) \right], \quad (38)$$

in agreement with the results derived in Ref. 7 by other means. We note that $\psi - \psi_0 = \bar{\psi}$ does indeed have the outgoing form specified at the beginning of this section, not only as $|x| \rightarrow \infty$ but for all x . For $x > 0$, we find

$$\psi(x) = (2\pi\hbar)^{-1/2} \frac{i\hbar p_0}{i\hbar p_0 - mc} \exp\left(\frac{ip_0x}{\hbar}\right), \quad (39)$$

from which we extract the transmission coefficient

$$T = \left| \frac{i\hbar p_0}{i\hbar p_0 - mc} \right|^2 = \frac{\hbar^2 p_0^2}{\hbar^2 p_0^2 + m^2 c^2}. \quad (40)$$

Similarly, for $x < 0$ we have

$$\psi(x) = (2\pi\hbar)^{-1/2} \left[\exp\left(\frac{ip_0x}{\hbar}\right) + \frac{mc}{i\hbar p_0 - mc} \exp\left(-\frac{ip_0x}{\hbar}\right) \right]. \quad (41)$$

From the coefficient of the second term (the reflected wave) we extract the reflection coefficient

$$R = \left| \frac{mc}{i\hbar p_0 - mc} \right|^2 = \frac{m^2 c^2}{\hbar^2 p_0^2 + m^2 c^2} = 1 - T. \quad (42)$$

As an intrinsically complex function it is difficult to

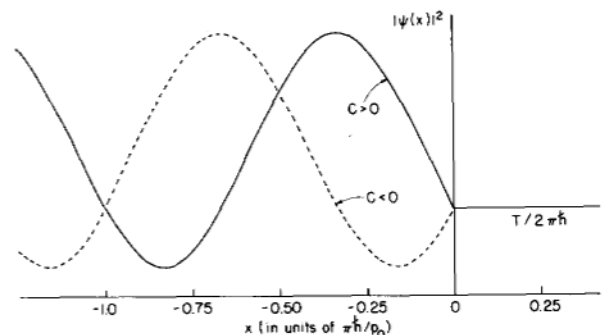


Fig. 3. Probability density $|\psi(x)|^2$ for the scattering problem in configuration space. For negative x the oscillatory nature of the solution is shown for both the repulsive ($c > 0$) and attractive ($c < 0$) cases, with the same value of $|c|$. The solutions for $x > 0$ are the same, with the values of $|c|$ and p_0 having been chosen to make the transmission coefficient $T = 3/4$ in the drawing.

sketch $\psi(x)$ in a meaningful way. In Fig. 3 we sketch the probability density $|\psi(x)|^2$. For $x > 0$ this is simply a constant $T/2\pi\hbar$, representing the transmitted pure plane wave of Eq. (39). For $x < 0$, $|\psi(x)|^2$ is oscillatory, showing interference between the incident and reflected waves (a standing wave).

Unfortunately, since $\phi(p)$, given by Eq. (37), is a complex generalized function, neither $\phi(p)$ nor $|\phi(p)|^2$ may be plotted (this is particularly obvious for the plane wave part ϕ_0). In fact, $|\phi(p)|^2$ does not even exist as a generalized function. Wave packets are essential for the physical interpretation in terms of probability densities in p space.

Before concluding, we wish to make a few remarks about the momentum space representation of the problem of scattering from an arbitrary potential, still confining ourselves to the one-dimensional case for simplicity. When we work in configuration space, we only know the general behavior of $\psi(x)$ in the asymptotic limit $|x| \rightarrow \infty$. In momentum space, however, we know the precise form of the solution, namely Eq. (32). Only the determination of the smooth function $f(p)$ remains; the boundary condition is satisfied identically by the presence of the Dirac factor. Surely this is a convenient place to begin an approximation scheme! If we insert Eq. (32) into Eq. (4) and make use of Eq. (23b), we arrive at an equation for $f(p)$:

$$f(p) = (2\pi\hbar)^{-1/2} \int_{-\infty}^{\infty} \tilde{V}(p-p') \phi(p') dp'. \quad (43)$$

Readers familiar with scattering theory will recognize this as a sort of p -representation version of the scattering amplitude formula $(\psi_0, V\psi)$ [for which we would have to set $p = p_0$ in Eq. (43)]. If we insert $\phi = \phi_0 + \bar{\phi}$ into Eq. (43), we have

$$f(p) = (2\pi\hbar)^{-1/2} [\tilde{V}(p-p_0) + \int_{-\infty}^{\infty} \tilde{V}(p-p') \bar{\phi}(p') dp'].$$

The familiar Born approximation follows from this equation by dropping the integral term containing $\bar{\phi}$ on the assumption that, in some sense, $|\bar{\phi}| \ll |\phi_0|$. In the case of the delta function potential, for instance, we get from Eqs. (7) and (44)

$$f(p)_{\text{Born}} = \lambda_{\text{Born}} = c/2\pi\hbar.$$

The exact result, Eq. (36), reduces to this value if $p_0 \gg m|c|/\hbar$. For a discussion (in x space) of the complete Born series for this potential, see Lapidus.⁷

In conclusion, we have illustrated at some length the solution of a problem in momentum space in order to show the nature of the techniques involved. For the bound state problem the particular potential used led to a solution perhaps even simpler than the x -space version, but in general one would need integral equation techniques, which are presumably less familiar than the differential equation methods usually employed. In the continuum case, the p -space treatment has the drawback of requiring the extensive use of generalized functions, which are not only less familiar than ordinary functions occurring in x space but appear to be even further removed from intuition than the unphysical plane waves

employed there. Nevertheless, the author believes that there are compensating advantages to the p -space approach and that it is both possible and worthwhile to develop the necessary intuition and facility in the handling of generalized functions.

APPENDIX

In this appendix we derive the key result used in Sec. IV of the text, namely that the outgoing wave condition of configuration space translates into the requirement that $\phi(p)$ contains the Dirac factor (26). Dirac's proof⁶ is for the three-dimensional case. Here we present the similar but simpler one-dimensional proof.

We begin with the general equation, Eq. (4). Inserting $\phi = \phi_0 + \bar{\phi}$ into Eq. (4) and using Eqs. (23b) and (24), we find the equation for the scattered part of the wave function $\bar{\phi}$:

$$(p_0^2 - p^2) \bar{\phi}(p) = g(p), \quad (\text{A. 1})$$

where $g(p)$ is a smooth function. [Actually, $g(p)$ depends upon ϕ . It is $2mf(p)$, where $f(p)$ is given in Eqs. (43) and (44). The smoothing action of the integrations give g the necessary smoothness in spite of singular structure of ϕ , and it is only the smoothness of g that we need.]

As argued following Eq. (29), Eq. (A.1) does not determine $\bar{\phi}$ uniquely. Rather, we can write the general solution

$$\bar{\phi}(p) = g(p) P[(p_0^2 - p^2)^{-1}] + A\delta(p+p_0) + B\delta(p-p_0), \quad (\text{A. 2})$$

since both of the last two terms are annihilated by $p_0^2 - p^2$ according to the identity (25). If we insert (A.2) into Eq. (3), we obtain the configuration space scattered part:

$$\psi(x) = (2\pi\hbar)^{-1/2} \left[A \exp\left(-\frac{ip_0x}{\hbar}\right) + B \exp\left(\frac{ip_0x}{\hbar}\right) + P \int_{-\infty}^{\infty} \frac{\exp(ipx/\hbar) g(p)}{p_0^2 - p^2} dp \right]. \quad (\text{A. 3})$$

We now wish to evaluate the last integral for large x . Using the partial-fraction decomposition

$$\frac{1}{p_0^2 - p^2} = \frac{1}{2p_0} \left[\frac{1}{p+p_0} - \frac{1}{p-p_0} \right],$$

we see that the integral decomposes into $1/2p_0$ times the difference of two integrals, each of the form

$$I(q) = \int_{-\infty}^{\infty} \frac{\exp(ipx/\hbar) g(p)}{p-q} dp, \quad (\text{A. 4})$$

with $q = -p_0$ and $q = p_0$ in the two cases. We may now write

$$I(q) = g(q) P \int_{-\infty}^{\infty} \frac{\exp(ipx/\hbar)}{p-q} dp + \int_{-\infty}^{\infty} \frac{\exp(ipx/\hbar)[g(p)-g(q)]}{p-q} dp. \quad (\text{A. 5})$$

$$P \int_{-\infty}^{\infty} \frac{\exp(ipx/\hbar)}{p-q} dp = \exp\left(\frac{iqx}{\hbar}\right) P \int_{-\infty}^{\infty} \frac{\exp[i(p-q)x/\hbar]}{p-q} dp = \exp(iqx/\hbar) \left[P \int_{-\infty}^{\infty} s^{-1} \cos(sx/\hbar) ds + i \int_{-\infty}^{\infty} s^{-1} \sin(sx/\hbar) ds \right]. \quad (\text{A. 6})$$

In the last expression we have used $s = p - q$ as the integration variable. The sine integral needs no principal value since the integrand is finite at $s = 0$. Now symmetry of the integrand shows that the principal value cosine integral vanishes. On the other hand, the sine integral is the well-known

$$\int_{-\infty}^{\infty} s^{-1} \sin(sx/\hbar) ds = \pi x/|x| \quad (\text{A. 7})$$

(i.e., $+\pi$ for $x > 0$, $-\pi$ for $x < 0$). Thus,

$$P \int_{-\infty}^{\infty} \frac{\exp(ipx/\hbar)}{p-q} dp = i\pi \frac{x}{|x|} \exp\left(\frac{iqx}{\hbar}\right) \quad (\text{A. 8})$$

and hence, as $|x| \rightarrow \infty$,

$$I(q) = i\pi(x/|x|) \exp(iqx/\hbar) g(q) + O(|x|^{-1}). \quad (\text{A. 9})$$

Correspondingly,

$$\bar{\psi}(x) = (2\pi\hbar)^{-1/2} \{A \exp(-ip_0 x/\hbar) + B \exp(ip_0 x/\hbar) + (i\pi x/2p_0 |x|) [g(-p_0) \exp(-ip_0 x/\hbar) - g(p_0) \exp(ip_0 x/\hbar)]\} + O(|x|^{-1}). \quad (\text{A. 10})$$

For $x > 0$ the outgoing wave moving to the right is $\exp(ip_0 x/\hbar)$. To cancel the incoming wave, $\exp(-ip_0 x/\hbar)$, we must have

$$A = -i\pi g(-p_0)/2p_0. \quad (\text{A. 11})$$

For $x < 0$ the outgoing wave moving to the left is $\exp(-ip_0 x/\hbar)$. To cancel the incoming wave, $\exp(ip_0 x/\hbar)$, we must have

$$B = -i\pi g(p_0)/2p_0. \quad (\text{A. 12})$$

Substituting these two results in (A.2), we have

The last integral needs no principal value since the integrand is smooth near $p = q$. As $|x| \rightarrow \infty$, this term will fall off at least as fast as $1/|x|$ (Riemann-Lebesgue theorem). The principal value integral on the right-hand side of (A.5) will dominate the second term as $|x| \rightarrow \infty$, and we must evaluate it carefully.

$$\bar{\phi}(p) = g(p) P[(p_0^2 - p^2)^{-1}] - (i\pi/2p_0) [g(-p_0) \delta(p+p_0) + g(p_0) \delta(p-p_0)]. \quad (\text{A. 13})$$

Because of the delta functions, we have

$$g(-p_0) \delta(p+p_0) = g(p) \delta(p+p_0), \quad (\text{A. 14})$$

$$g(p_0) \delta(p-p_0) = g(p) \delta(p-p_0).$$

We may then factor out the $g(p)$ and use the identity (33) to obtain the final result:

$$\begin{aligned} \phi(p) &= g(p) \{ P[(p_0^2 - p^2)^{-1}] - i\pi \delta(p_0^2 - p^2) \} \\ &= f(p) \{ P[(E - p^2/2m)^{-1}] - i\pi \delta(E - p^2/2m) \} \end{aligned} \quad (\text{A. 15})$$

with $f(p) = g(p)/2m$, a smooth function.

¹A. Messiah, *Quantum Mechanics* (Wiley, New York, 1962).

²E. Merzbacher, *Quantum Mechanics* (Wiley, New York, 1970), 2nd ed.

³See Ref. 1, pp. 326-329; Ref. 2, p. 140 ff.

⁴Notable exceptions are the simple harmonic oscillator, which is essentially the same in x or p space, and the hydrogen atom bound state problem, which has been frequently discussed in this Journal, presumably because the momentum representation exhibits the hidden symmetry.

⁵See Ref. 1, p. 820 ff; Ref. 2, p. 223 ff.

⁶P. A. M. Dirac, *Quantum Mechanics* (Oxford University, London, 1958), 4th ed., Sec. 50.

⁷G. Baym, *Lectures on Quantum Mechanics* (Benjamin, New York, 1965), pp. 113-115; D. S. Saxon, *Elementary Quantum Mechanics* (Holden-Day, San Francisco, 1968), pp. 410-411; P. M. Morse and H. Feshbach, *Methods of Theoretical Physics* (McGraw-Hill, New York, 1953), p. 1644. A useful mathematical reference is B. Friedman, *Principles and Techniques of Applied Mathematics* (Wiley, New York, 1957), p. 243 ff. Various aspects of the delta function potential in x space have been considered in this Journal. See, for example, P. B. James, *Am. J. Phys.* **38**, 1319 (1970); I. R. Lapidus, *Am. J. Phys.* **37**, 930 (1969); **37**, 1064 (1969).

⁸Reference 2, Eq. (6.25).

⁹Reference 1, Eq. (A.19).

¹⁰Reference 1, Eq. (A.15e); Ref. 2, Eq. (6.18).

¹¹Reference 1, Eq. (A.18); Ref. 2, Eq. (6.16).